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Identifying and Preventing ESP Failures Resulting from Variable Speed Drive Induced Power Quality Issues

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Abstract

The effect of Variable Speed Drive (VSD) waveforms on electrical motors is generally well understood; however, typical industrial solutions fail to adequately protect specialized ESP systems, as ESP motors cannot be drive-duty rated. This paper examines VSD/ESP systems, discusses how switching transients and high frequency content may manifest themselves via the carrier frequency of the VSD, addresses the harm those transients may cause to ESP systems, and provides solutions that can mitigate related ESP failures.

Dismantle, inspection, and failure analysis of the VSD-powered ESP equipment indicated that there was an underlying electrical issue with these systems, thus comprehensive power quality studies were conducted. Data was collected on unfiltered VSD outputs, after the installation of Sine Wave (line to line) filters, and after the installation of Common mode (line to ground) filters. Current and voltage data was collected using Fluke 1750 power quality analyzers, as well as Tektronix THS3024 and Yokogawa PX8000 oscilloscopes with appropriate high frequency test probes. Spectral analysis (also known as harmonic analysis, or frequency domain analysis) was then performed on the collected data, from which corrective actions were determined.

Spectral analysis indicated that even on sine wave filtered systems, there was always significant common mode current present. While these currents contribute relatively little to Total Harmonic Distortion between 20th and 73rd harmonic (dependent on carrier frequency), it was found if their individual resultant amperage was greater than only 0.3 amps L-G, ESP failures would occur. Failures were caused by bearing fluting; feedthrough and penetrator arcing; stator end turn failures; skin effect heating of cables; and MLE, pothead, or splice insulation failures. Due to the specialized application of ESP's, motor design changes have not been readily implemented to improve their resistance to high frequency electrical content. Therefore, the only reasonable solution is to improve VSD output power quality to conform to IEEE 519-2014 specifications in the high frequency realm, with special attention paid to reducing harmonic content beyond the 20th harmonic. The installation of well-designed sine wave and common mode filters provides the most cost effective means to correct poor power quality in ESP systems, thus both types of filters were installed in the case presented. Adding these electrical filters resulted in the ESP mean time between failure to typically double, but most importantly, resulting failure modes were attributed to other non-electrical root causes.

During testing, an inspection of ESP wellhead cable utilizing airborne ultrasonic scanning tools was performed. The operator detected the distinct sound of an inverter over 1000' away from the VSD that was powering the ESP. As filters were installed, the sound decreased and was ultimately not detectable. It is believed that with enough research, the use of airborne ultrasound may provide a novel, non-invasive means to detect harmful high frequency electrical signals in ESP systems.

Introduction

Over the last decade, the use of VSD's in ESP applications has increased due to the flexibility in production capacity they offer. Additionally, due to a VSD's ability to control electrical outputs, motor inrush currents can be limited such that excessive heating of the pothead (and the subsequent related failures) cannot result. These benefits have resulted in operators shifting from "Starting switchboards" to VSD's for many ESP applications. However, with this shift have come some unintended consequences, most commonly indicated by substantially shorter ESP run life, and new failure modes emerging throughout the ESP system. Recent research has proven that electrical discharge phenomena will result in bearing failures within ESP's operating on pulse width modulating (PWM) VSD's without functional dV/dt filters; however little was mentioned in mitigating the common mode voltage which was correctly identified as a source of harmful shaft currents [1]. This paper is designed to provide a fundamental understanding of the science, equipment and analysis techniques required to troubleshoot and solve ESP system failures that stem from insufficient power quality. Our goal is to provide production engineers the knowledge and tools required to ensure their equipment maintains the highest levels of performance and reliability possible. The situation analyzed in this paper covers ESP's used in Geothermal applications. Geothermal ESP pump set depths are typically between 500' and 1800', with ambient temperatures of 270°F to 330°F and flow rates in excess of 30,000 BPD, per well.

Technical Background Information

While industry nomenclature often defines the ESP system as starting at the wellhead penetrator, the exclusion of required surface equipment such as VSD's, controls, transformers, and electrical supply results in an incomplete picture of system reliability and the sophisticated interplay of the various components. Therefore, this paper deviates from industry norms, and defines the system as shown in [Figure 1](#), which includes the input power to the VSD. The following sections provide background on the electrical and mechanical fundamentals required to understand the entirety of the ESP system when powered on grid or gen-set tied VSD/ESP systems.

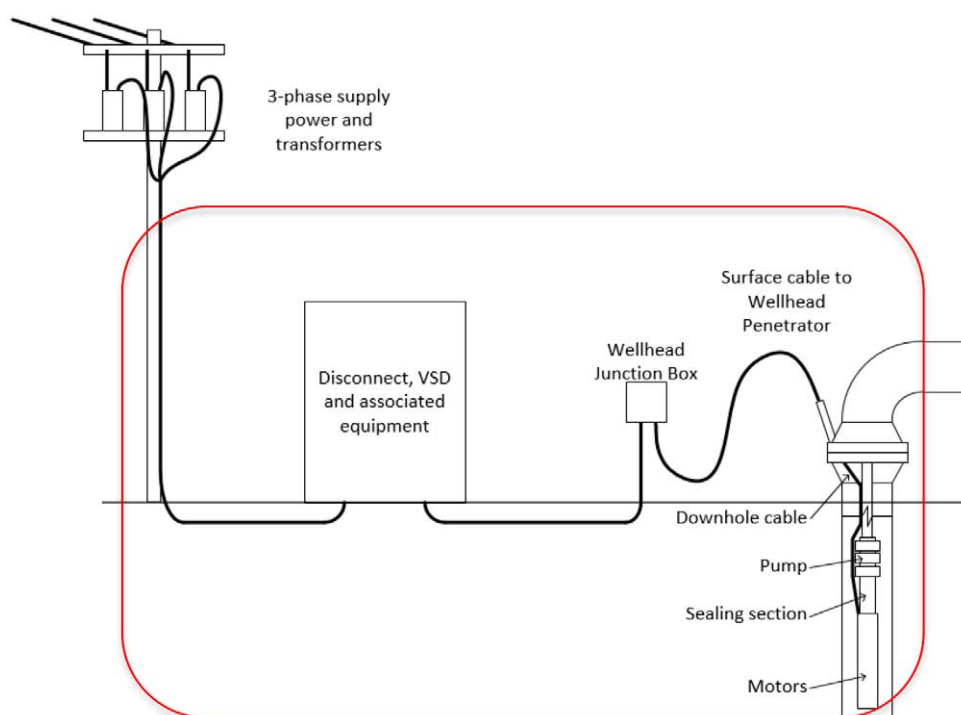


Figure 1—Typical ESP system outlined in red. Note this includes supply power, intermediary electrical equipment and cables, and all downhole equipment.

General Discussion of Three-Phase Electric Power

Three-phase electric power is composed of three separate sinusoidal electric signals, each 120° out of phase with one another, and is the common form of alternating current electricity (AC) used in power distribution and transmission. A typical example of a balanced three-phase sinusoidal plot is shown in Figure 2. These sinewave plots can be decomposed into simple vector diagrams, which show the RMS voltage or current of each phase as shown in Figure 3. Furthermore, we can extend upon this concept to show a simplified magnetic field diagram as shown in Figure 4. By inspection, we can see that the three phases, regardless of the graphic we choose, are balanced in magnitude, and due to their phase relationships, cancel each other out perfectly. The right vector diagram in Figure 3 shows this most clearly, as the three-phase vector sum clearly equals zero.

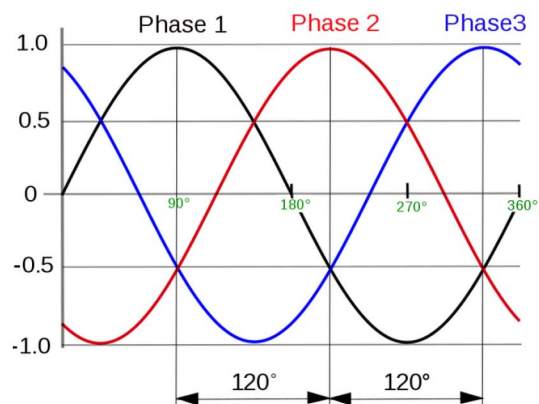


Figure 2—An example of an ideal 3-phase sinewave signal

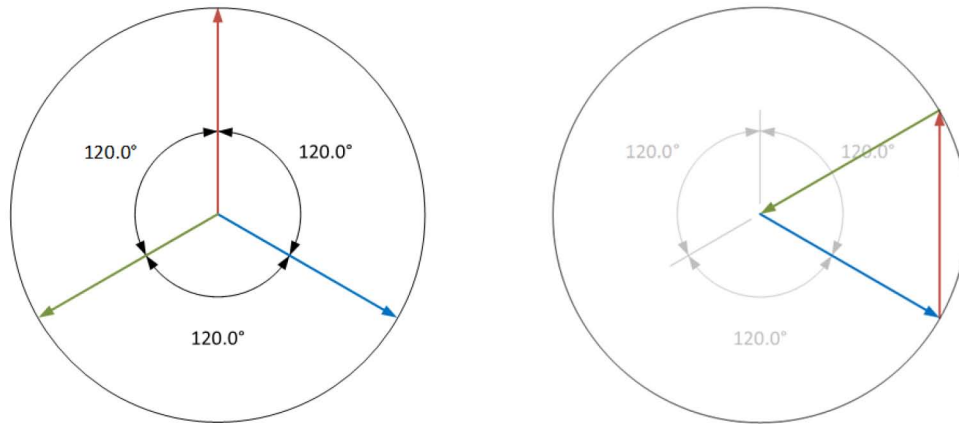


Figure 3—Balanced three phase vector diagrams. Left showing the phasors, right showing their resultant sum

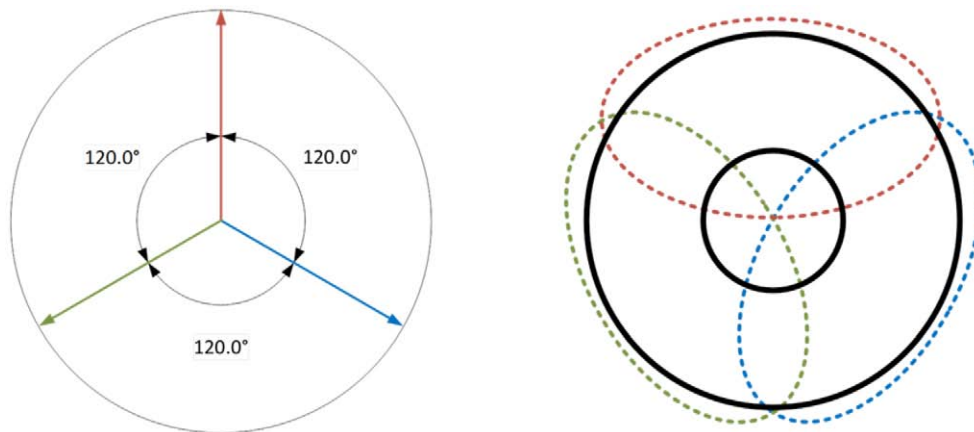


Figure 4—Relationship between balanced three phase currents (left) and magnetic fields (right)

General Discussion of AC Induction Motors

An AC motor is simply a type of rotating transformer. A cross-sectional schematic of a typical surface AC motor is shown in Figure 5, showing the arrangement of the stator, bearings, shaft and rotor. Note that a single rotor spins within the stator, and the two are isolated by an air gap. As the currents alternate in the motor windings within the stator, the magnetic field strengths change as well. These field changes are what induce electric currents within the rotor; currents which thereby create their own magnetic fields, this phenomenon is why most AC motors are called "induction" motors. The interaction between these magnetic fields causes the torque which makes the motor shaft to rotate. When the three motor phase currents are well balanced, the magnetic fields are balanced as well, and no electrical potential exists between the various components of the motor. In this case the motor requires no special design considerations, and it can run well so long as the voltage/currents remain balanced.

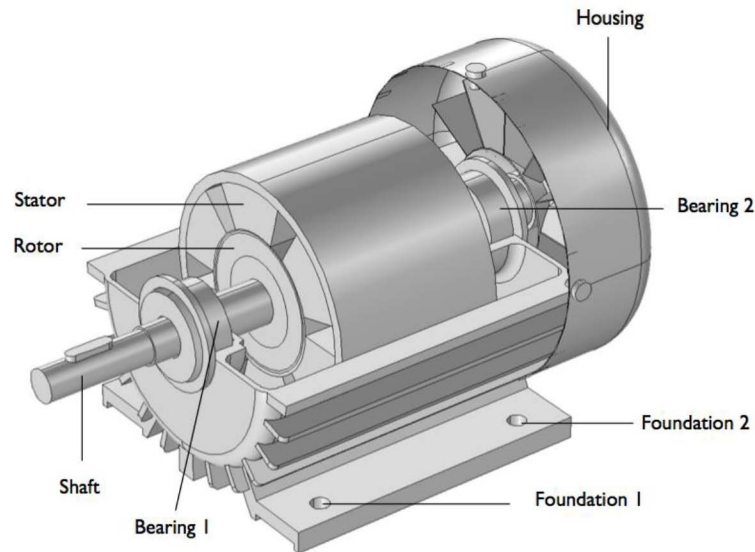


Figure 5—A cross section of a typical surface induction motor [2]

General Discussion of VSD's

VSD's and all related inverter architectures operate under the same basic principles. Incoming three-phase power is converted (rectified) within the drive to a DC electric signal. This DC power is then sent through a set of switches (inverter) that quickly turn on and off in order to simulate an AC electrical signal of the appropriate amplitude, frequency and other electrical attributes. Figure 6, shows the key components and simplified topology of a basic rectifier, DC link, and inverter. Williams, Baillie and Shipp, in their 2003 SPE paper "Understanding VSDs with ESPs – A Practical Checklist" provide an excellent overview of VSD's and their functions; therefore the topic shall not be entirely restated here (readers are strongly encouraged to reference this paper, however) [2]. Nonetheless, it is critical to understand the inverter section of the VSD, and the result it has on power quality on the output of the VSD – the power provided to the ESP.

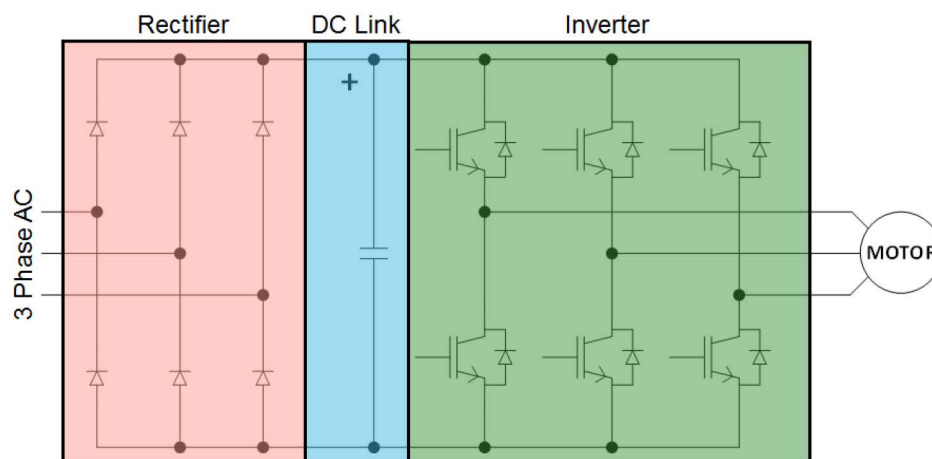


Figure 6—A simplified VSD topology showing major sections of the drive.

Regardless of the specific drive topology, nearly all inverter designs utilize a type of high frequency switch to simulate an AC waveform. This high frequency switching does not create a perfectly smooth output waveform, rather it creates a "stair stepped" waveform that approximates the shape of the desired sinewave. Figure 7 shows typical voltage and current waveforms for the three most common drive topologies used with ESP's. Of particular note, are the shapes of the current waveforms; aggressive and jagged in six-step

topology, saw-toothed in a standard PMW topology, and smoother (but also saw toothed) in the multilevel PWM topology. All known industrial drives will generate signals such as these at the output of the drive inverter sections, without exception. The spikes on these signals are the result of the switching process, called commutating transients (transients for short), and are artifacts of the drive design and the inductive reactance of the entire system [2]. These transients are a significant contributing source of damage within non-inverter rated AC motors, which includes ESP's.

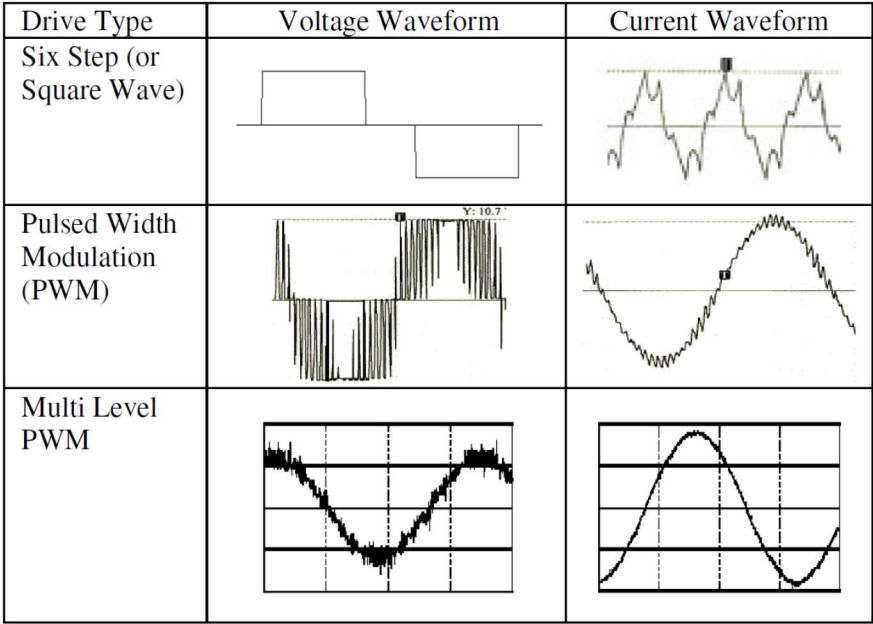


Figure 7—Various waveforms based on drive architecture [2]

General Discussion of Harmonics and Spectral Analysis

When one examines any simple sinewave, it is clear that wave can generally be described mathematically by the equation:

$$y(t) = A \sin(2\pi ft + \varphi)$$

Where: A is the amplitude, f is the ordinary frequency, and φ is the phase of the signal in radians.

This basic equation works well when analyzing simple sinusoidal waveforms, but does not suffice with complicated signals such as those generated by VSD's. Therefore, a set of mathematical algorithms called the "Fast Fourier Transform" (FFT) are utilized to break complicated signals down into a sum of discrete sinusoids, as shown in figure 8. The amplitudes of these sinusoids are listed in order of their individual frequencies, or are alternatively listed in order as multiples of the fundamental frequency. Multiples of the fundamental frequency are called "harmonics" and are critical to understanding how "clean" the electrical waveform is. For example, on a system with a 60Hz fundamental (such as in North America), 120Hz would be the second harmonic, and 4200Hz would be the 70th harmonic. Similarly, if a VSD is operating at a 57.25Hz fundamental, the second harmonic would be 114.5Hz, and the 70th harmonic would be 4007.5Hz. When investigating high frequency electrical signals that may be hidden within complicated waveforms, spectral analysis allows for the identification of the source of the specific frequency by correlating what's found to the forcing frequencies of the system design. Typical PWM VSD's have carrier frequencies between 800Hz and 20,000Hz, however in the ESP industry, most carriers are between 1800Hz to 5000Hz [2]. Additionally, lower order harmonics tend to distort the waveform significantly, and higher order harmonics tend to "ride" the fundamental. Six-step drives create considerable lower order harmonics due to their square wave voltage output; hence they create aggressive current wave forms as was previously shown. Multilevel

PWM VSD's tend not to have significant low order harmonics, but beyond the 20th harmonic may generate significant levels of distortion.

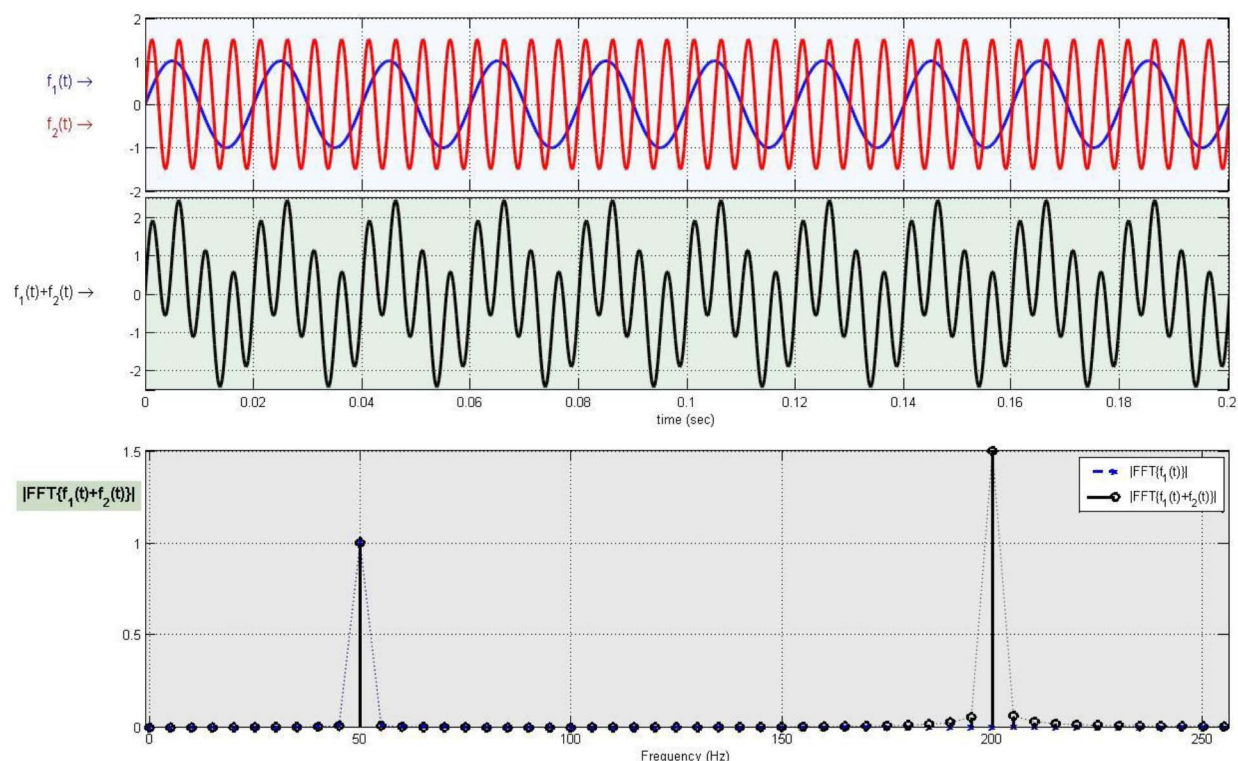


Figure 8—The top plot shows two discrete sinewaves $f_1(t)$ and $f_2(t)$. The middle plot shows the sum of $f_1(t)$ and $f_2(t)$. The bottom shows the resultant spectrum after performing an FFT of the complex waveform $f_1(t) + f_2(t)$ [3].

Finally, a simple way to quantify how "clean" or "dirty" a signal is, is via the equation for "Total Harmonic Distortion (THD)." The general equation for THD is:

$$THD_F = \frac{\sqrt{V_2^2 + V_3^2 + \dots + V_n^2}}{V_1}$$

Where: V_1 is the RMS value of the fundamental, and V_2 to V_n are the RMS values of the n^{th} harmonic analyzed.

The highest harmonic, n , evaluated by the user is a function of the equipment used to collect the data and the user's interest. For typical utility power quality analysis, THD calculations generally stop at the 20th harmonic, but for VSD's they may need to be calculated up to or beyond the 100th harmonic, depending upon the carrier frequency of the VSD. Therefore caution must be exercised when discussing THD, such that the number of harmonics included in the calculation is clearly stated. This is because the calculated THD up to the 20th harmonic can (and often is) quite different than THD calculated to the 100th. To assume that values beyond a certain harmonic are of no value could be a costly mistake.

General Discussion of VSD-Derived Three-Phase Characteristics and Anomalies

Due to the high speed switching of VSD's, the wave forms they create are not pure sine waves, but are a composite of numerous sinusoids that approximate a single pure sine wave, as shown previously in [figure 7](#). By utilizing an oscilloscope and high frequency probes, it becomes possible to inspect the specific commutating transients as shown in [figure 9](#), which is a typical PWM waveform. Of note, is that the peaks of the transient are nearly $\sim 1000V$ higher than the mean for that pulse. Therefore, if we examine the three

phase relationships of such a waveform, we can generate diagrams similar to those mentioned in section 3.1. By sampling the discrete 3 phase values of any PWM waveform, we see that the phase values no longer sum to zero, resulting in a phase imbalance in the system as shown in figure 10. The resulting magnetic fields are therefore imbalanced as well, thus the system generates a "zero sequence" current, which restores balance to the system as shown by the bold black arrow in figure 10. On a three-phase system with a neutral, this would be measured as a neutral current. However, on the systems without a neutral such as ESP's, it is manifested by additional load between the phases, and electrical potential between the phases and ground.

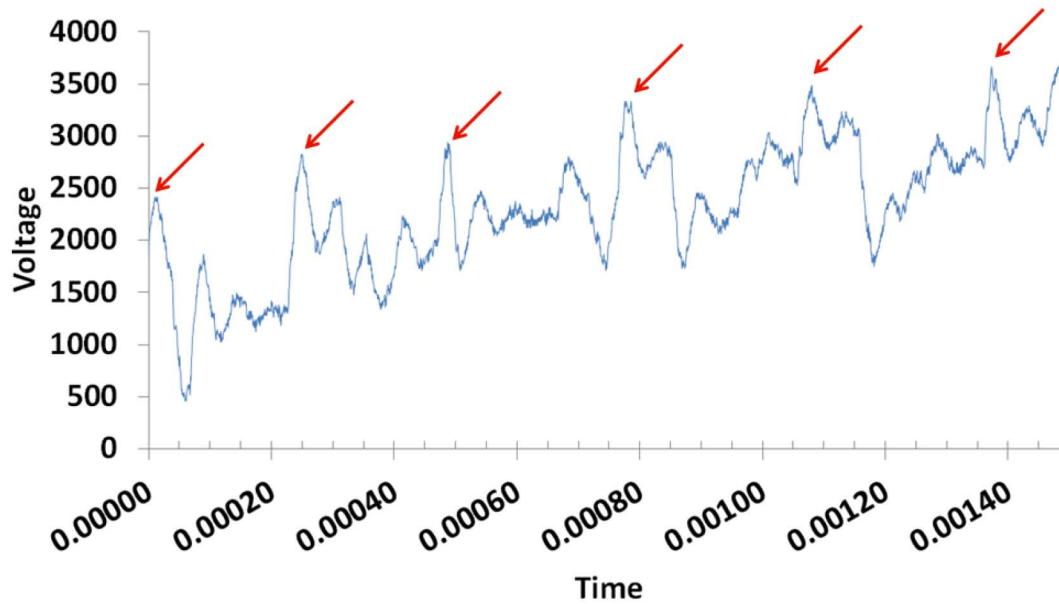


Figure 9—A zoomed in view of the voltage wave form, sampled at 75MHz, showing the primary commutating transients highlighted by the red arrows.

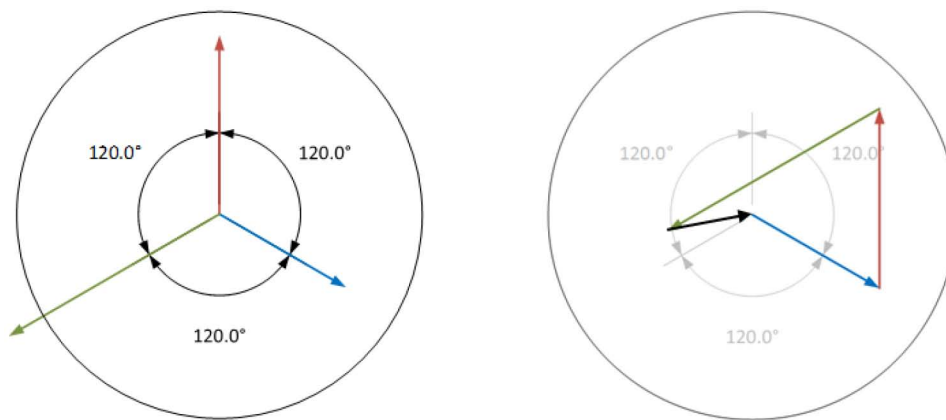


Figure 10—An imbalanced three phase vector diagrams. The left graphic is showing the phasors, the right showing their resultant sum. The black arrow shows the zero sequence voltage/current required to balance the system.

It is critical to note that the high frequency transients that are created by VSD's do not behave the same way 50/60Hz fundamental frequencies do. In particular, high frequency electrical signals travel on the very outer skin of the conductor, and do not pass through obstructions well, but instead tend to reflect back up the conductor. Like any wave, these reflections can be added to other waveforms on the conductor, thus possibly creating amplitudes double those measured directly off the VSD output. These traits of high frequency electrical signals therefore have the following consequences:

Table 1—The relationship between high frequency traits and their effects on the ESP system

Trait	Consequence
Reflects at splices and at tight conductor radii	Stator, Splice and Cable failures due to summing of the reflected commutating transients
Travels over conductor, not in it	Skin effect heating, degraded insulation performance
Adds to total system current, but does no work	Increased motor temperature and higher electrical costs
Capacitive coupling increases with frequency	Increased electrical noise in the system impacting sensors and nearby instruments
Insulation strength decreases with frequency	Motor shorts, bearing “fluting” and “frosting”

The final consequence of all of these issues is ultimately the same; reduced life and reliability for equipment not specifically designed to operate on VSD supplied power. Unfortunately, the severity of these issues is highly dependent on the specific system design. Identical ESP systems, running at different speeds may manifest substantially different mean times between failures. Additionally, cable lengths between the VSD and the motor contribute significantly to potential harm from VSD's; with longer cable lengths being worse than shorter cable lengths. This factor is so significant, that the US Department of Energy acknowledges that damaging reflected waves can become problematic at cable lengths exceeding 15 feet [4]. Considering many ESP's supply cables are in excess of 5,000 feet, the issue becomes even more critical in those applications. To ultimately understand why similar systems perform differently, the harmonics generated by the VSD at various speeds and loads need to be investigated.

General Discussion of Inverter-Rated AC Surface Motor Systems

In situations where a standard (non-inverter duty) AC induction motor is powered by a VSD, failure rates will almost certainly increase, as these motors do not have design features that mitigate line to line (L-L) and line to ground (L-G, or common mode) currents. Thus there are several design differences between inverter rated AC motors and standard AC motors such as the choice in magnet wire, bearing isolation systems, and added grounding rings. Other modifications to the motors typically include improved cooling, and higher temperature ratings on insulation. Through basic inspection of an inverter duty and non-inverter duty AC motor, almost none of these modifications are obvious. However, all of these design changes are included in order to diminish or entirely mitigate the consequences of the VSD induced electrical transients mentioned in section 3.5. Without the inclusion of these features, numerous failure modes will manifest within the system. Commutating transients and the reflected high frequency signals will place electrical insulation under constant attack, especially at the stator end turns. Circulating currents will arc across bearings, resulting in a fluted or frosted appearance of the sleeve/journal, and lead ultimately to premature motor bearing failure. Additionally, in cases where the VSD is a 6-step drive or similar, the low order harmonics will generate pulsing torques in the motors which can result in shaft failures within the motor and the driven machine as indicated by 45° shaft fractures. Finally, the high frequency electrical content results in skin effect heating which breaks down motor insulation, which further degrades expected motor life. On well-designed VSD driven systems, even the motor cable differs from the non-VSD counterpart. The cables have extra shielding to prevent noise on nearby conductors, have higher rated insulation to contain the high frequency signals, higher conductor strand counts, and enhanced grounding capabilities. Without these features, the cable life will also be shortened, and nearby systems will have significant voltages induced upon them; possibly damaging them as well [5]. Needless to say, a properly design inverter-rated AC surface motor system differs significantly from its non-VSD equivalent. So we are forced to ask, how do ESP's compare?

General Discussion of ESP motor design and VSD's

While ESP motors are indeed AC induction motors, with all the same basic components and operating principles, their similarity to surface motors stops there. The most striking difference is the dramatic difference in aspect ratio. A typical 1500HP/1100kW surface motor will have a stator approximately 30" wide and 36" long with an aspect ratio of ~ 1.2 , with a single rotor section. An ESP of the same horsepower will actually be tandem motors less than 9" in diameter and with a total motor section approximately 60' long for an aspect ratio ~ 81.8 . This dramatic design difference is for no other purpose than to fit the entire system down a well bore, and it results in numerous design differences from industrial surface motors. Since the motor is dramatically longer, the rotor is broken up into numerous segments each supported by bearings that ride internal to the stator, thus a 1500HP tandem motor has approximately 40 bearings. With so many bearings internal to the motor, coupled with high environmental pressures from the well fluids, ESP's are oil filled. This oil fill replaces the airgap found in surface motors, and serves as a coolant, dielectric and lubricant for the motor. As a final major difference from surface motors, the small motor diameter of ESP's results in tight stator winding end turns which impacts electrical characteristics of the motor as well.

Due to space constraints in the wellbore, ESP's do not commonly exhibit the features typically found in inverter-rated surface motors, such as insulated bearings, shaft grounding rings, and specially designed stators. Additionally, most designs in service today are some variation of those originally designed before the use of VSD's became prevalent, thus other design considerations such as iron-copper ratios are not optimized for VSD applications. In short, ESP's do not include the design characteristics one would expect to see for an AC induction motor for VSD applications. In fact, certain characteristics are common to most ESP designs, such as tight stator end turn radii, which are actually a worst case scenario for VSD applications, as the high frequency content of the wave form has a tendency to reflect at the end turns causing excessive heating and voltage spikes near that location. Wellbore diameter constraints make it difficult or impossible to design ESPs as inverter-duty motors, and the issues are only exacerbated as the wellbore and ESP diameters are reduced. Thus other means of motor protection at the surface should be realized as a lower cost means of improving motor life.

History

In 2015 six geothermal water wells located at two separate fields in the western United States began experiencing short run times after commissioning their VSD's. ESP pump set depths for these fields are approximately 1500', with surface cables ranging from 1000' to 4700' from the well head. Due to the geothermal environment, downhole ambient temperatures range from 310°F to 330°F. These ESP systems pump the hot geothermal fluids to the surface at rates in excess of 100,000 BPD. These wells are pumped utilizing 1000hp-1500hp ESP (tandem, derated, 750HP units), powered by 5kV VSD's utilizing multilevel pulse-width modulating drive topologies. The multilevel PWM drive topology is generally considered to have among the lowest dV/dt stresses (line to line), lowest common mode voltages (line to ground), and least waveform distortion of common industrial drives, however at an increased investment for the drive [7]. In all cases, these wells are powered directly off the electrical bussing of the power plant they serve, thus input power quality to the VSD is closely monitored and has very low harmonic content. Initial evidence of problems arose 178 days after installation of the first ESP, when a motor failed on a ground fault. Teardown and failure analysis concluded that the cause of the failure was due to water intrusion through a loose fitting, however the bearings showed unusual wear patterns, reminiscent of sandblasting on the sleeve O.D. and bearing I.D., and a copper coloring across most surfaces as shown in [figure 11](#). There was no evidence of unusual mechanical wear on these bearings; the sand blasted appearance was asymmetrical, uneven, and constrained to areas of the sleeve directly contained by the bearing. This first instance of discoloration and pitting of the bearings was attributed to water intrusion inside the motor and subsequent galvanic corrosion. However, 274 days after the replacement ESP was commissioned, the same well failed on instantaneous

over current. Teardown and failure analysis of the second ESP system found the rotor mechanically locked, and the motors exhibited the same bearing anomalies as the first instance (see [figure 12](#)), however in this case there was no water found within the motor. The motor oil had fine brass colored particles suspended throughout, and the oil failed dielectric tests. These findings suggested that galvanic corrosion was unlikely to be the root cause of this failure; thus a technical investigation was launched by the authors to understand and eliminate this failure mode from the ESP fleet. Since these ESP's had been used for years in similar non-VSD applications without this type of failure, the VSD became the most likely contributing factor towards the equipment failures. Over the following 18 months and across 10 separate ESP system failures, the authors performed detailed power quality studies of the ESP systems and performed teardown and inspections of every failed ESP with the motor manufacturers to determine the root cause of failure. The following sections of this report detail the data collection and key results of these studies such that others can understand the techniques required to diagnose and correct VSD output power quality related failures.



Figure 11—Bearing from first failure



Figure 12—Bearing segments from second failure

Method

Motor Teardown and Inspections

Each failed motor and cable was dismantled at the manufacturers' facility for inspection and root cause analysis. In total, 20 individual motors were disassembled and inspected. Oil samples were collected as each major section of the motor was broken open. They were visually inspected, and tested for dielectric strength. The rotor strings were removed from the motors, and each of the bearings and individual rotors were removed from the shaft. The components were visually inspected, and unusual findings were photo documented along with a written report.

Laboratory Analysis

Four bearing samples from 4 different motors at various stages of failure were collected for 3rd party laboratory analysis, as shown in figure 13. These lab tests included Stereomicroscopy and Scanning Electron Microscopy (SEM) of the bearing surfaces, measurements of unusual structures that were identified, Fourier Transform Infrared Spectroscopy (FTIR) of the oil collected, chemical analysis of the bearing and sleeve materials via energy dispersive x-ray spectroscopy (EDS), and metallographic preparation of the samples in order to identify the cause of fractures. Additionally, virgin oil samples were provided to the lab for comparison to the oil recovered directly from failed motors. These tests were performed before and after cleaning of the bearing samples, such that any foreign materials could be identified. Cleaning was completed using an ultrasonic acetone bath, which removes oil and foreign material deposits from the surface of the bearing. Any materials embedded in the surface due to mechanical forces would remain. After inspection of the bearings was completed, representative specimens were prepared in cross-sectional metallurgical mounts. These mounts were again examined in an unetched condition in order to identify surface irregularities or defects. They were then etched in order to reveal material structures and aid in identifying any underlying metallurgical issues.



Figure 13—Bearings sent to laboratory for analysis

Electrical testing

Measurement of medium voltage waveforms (over 600Vac) involves safety and equipment protection considerations that differ from low voltage systems. In particular, most power quality analyzers and oscilloscopes are only rated for 600V, so special equipment must be acquired.

For the data collected, the authors have successfully used the following equipment for various VSD output power quality measurements:

Table 2—List of equipment used in the course of the power quality studies

Equipment Make-Model	Function	Sampling Rate	FFT Range
Yokogawa PX8000	“Powerscope” Oscilloscope	100MHz	1-500 th Harmonic
Tektronix THS3024	Handheld Oscilloscope	200MHz	1-50 th Harmonic
Fluke 1750/V3	Power Quality Recorder	5MHz	1-50 th Harmonic
Textronix P6015A	High Voltage 40kV-p Probe	75Mhz	n/a
RPM-3014	0.01A-40A Current Transformer	5kHz	n/a
RPM-3110	2A-100A Current Transformer	7kHz	n/a
FLU-I2000FLEX	20A-2000A Current Probe	20kHz	n/a
AEMC-SR661	100mA-2000A Current Probe	100kHz	n/a

The carrier frequency for these particular VSD's ranged between 2.4kHz and 3.6kHz, thus satisfaction of the Nyquist theorem required sampling rates approximately twice those values. Primary current and voltage (line to line, line to ground) measurements were taken with a Fluke 1750 and a Yokogawa PX8000. However, due to the connections available on the Fluke 1750, industrial potential transformers (PT) were utilized to collect voltage waveforms. Since these PT's are designed for high accuracy at 60Hz, they attenuate around 2000Hz, thus under representing very high frequency waveform components. Thus, the Yokogawa PX8000 Powerscope with Textronix P6015A voltage probes were used to measure high frequency voltage waveforms and to aid in validating the measurements taken with the Fluke 1750. Various current transformers were used according to actual motor currents and the measurements needs. For example, when measuring the currents through the Common Mode Filter, the RPM-3014 was used in order to improve measurement accuracy.

Measurements were taken at the wellhead junction box, approximately 1000' from the VSD, and 1700' from the motor. Measurements of both voltage and current waveforms from the multilevel medium voltage VSD and 1500hp ESP were taken in the three following system test configurations: Test 1, with no filters; Test 2, with sinewave filter; and Test 3, with sinewave and common mode filters. Additionally, measurements were taken during Test 3 at the output of the common mode filter in order to quantify the high frequency electrical content that it shunted to ground.

The Application of IEEE 519-2014 on VSD/ESP Systems

IEEE 519-2014 is a utility power quality standard which outlines the voltage and current harmonic limits up to the 50th harmonic. This standard was developed to increase utility equipment reliability as more non-linear loads were put on their distribution networks (such as VSD's). Since ESP's require electricity, and empirical evidence suggests they run best on line power, the authors have determined that this standard should be loosely implemented to the output power quality of the VSD to ensure the ESP electrical system is provided good power quality. This was deemed the most logical solution, as there is no accepted national or international standard regarding the power quality delivered from a VSD to its load. Tables 3 and 4 show relative limits for voltage and current harmonics, with the medium voltage sections highlighted in red.

Table 3—Voltage distortion limits per IEEE 519-2014 [8]

Bus voltage V at PCC	Individual harmonic (%)	Total harmonic distortion THD (%)
$V \leq 1.0$ kV	5.0	8.0
$1 \text{ kV} < V \leq 69$ kV	3.0	5.0
$69 \text{ kV} < V \leq 161$ kV	1.5	2.5
$161 \text{ kV} < V$	1.0	1.5 ^a

^aHigh-voltage systems can have up to 2.0% THD where the cause is an HVDC terminal whose effects will have attenuated at points in the network where future users may be connected.

Table 4—Current distortion limits for systems rated 120V through 69kV per IEEE 519-2014 [8]

Maximum harmonic current distortion in percent of I_L						
Individual harmonic order (odd harmonics) ^{a, b}						
I_{sc}/I_L	$3 \leq h < 11$	$11 \leq h < 17$	$17 \leq h < 23$	$23 \leq h < 35$	$35 \leq h \leq 50$	TDD
$< 20^c$	4.0	2.0	1.5	0.6	0.3	5.0
$20 < 50$	7.0	3.5	2.5	1.0	0.5	8.0
$50 < 100$	10.0	4.5	4.0	1.5	0.7	12.0
$100 < 1000$	12.0	5.5	5.0	2.0	1.0	15.0
> 1000	15.0	7.0	6.0	2.5	1.4	20.0

^aEven harmonics are limited to 25% of the odd harmonic limits above.

^bCurrent distortions that result in a dc offset, e.g., half-wave converters, are not allowed.

^cAll power generation equipment is limited to these values of current distortion, regardless of actual I_{sc}/I_L .

where

I_{sc} = maximum short-circuit current at PCC

I_L = maximum demand load current (fundamental frequency component)
at the PCC under normal load operating conditions

Results

Motor Teardown and Inspection

In all 20 individual motors inspected (upper tandem and lower tandem individually), the sleeves exhibited copper colored plating over the normally bright brass colored sleeve material. Additionally, the motor shaft often had similar copper colored tones, and the end turns of the rotors were plated in a similar manner to the sleeves. In all cases, bearings within the motors exhibited areas with a frosted appearance, with several total bearing failures located towards the middle of the rotor string. Moving from the middle of the motor, damage reduced as the bearings were progressively closer to the head bushings, however bearings towards the bottom of the motor exhibited more damage than the upper bearings. Often, the upper bearings

only exhibited minor radial scoring, with no other damage aside from the copper plating and minor areas of frosting. Additionally, all inspected motors had oil that was dark in color, and had a metallic "glitter" throughout. This "glitter" was composed of fine brass colored particles suspended throughout, with the highest concentration of particles taken from the oil at the bottom of the motors. These particles made the oil feel gritty and rough to the hand. In all cases, the oil failed factory dielectric strength tests. On all of the tandem systems, one of the tandem motors had more significant damage than the other, as exhibited by rotors that were unable to rotate, and extremely difficult to remove from the stator. Often, disassembly of the rotor string required the bearings to be forcibly removed using a hammer and chisel as they were mechanically locked to the shaft. This locked rotor condition was the most typical finding across the majority of the motors inspected during these series of failures, however there were three notable exceptions. In one case a stator end turn of an upper tandem motor failed, resulting in a melted region approximately $\frac{1}{2}$ " in diameter. In a separate case, an ESP system had a cable failure near the MLE at a downhole cable splice. This section of the cable failed numerous insulation resistance tests; however the relatively new tandem motors passed all mechanical and electrical integrity checks. For several meters above the splice, the insulation was loose and no longer adhering to the conductor. In the third case, several bearings had failed but the rotor could still turn; however the stator in this case failed insulation testing.

Bearing Lab Inspection

Oil analysis indicated that the used oil, virgin oil, and oils removed from the surface of the bearings were in all cases aliphatic hydrocarbon oils of the same type. No water was found in any samples. The bearings samples all indicated a small amount of carboxylic ester on the surface. No other unusual findings were noted.

Material analysis of the bearings indicated that the bearings and sleeves were of expected alloys and hardness (52 Rockwell B for the leaded red brass sleeve, and 42-43 Rockwell C for the stellite cobalt-base alloy bearing). There were no obvious defects in the material grain structure, nor were there any foreign contaminants identified. Except in the most extreme of cases, no mechanical damage was noted aside from light scoring and smudging. A pitted and frosted appearance was noted in specific areas of the bearing and sleeve; these areas of interest were then examined more closely.

The results of this detailed inspection were performed initially under 50x magnification, with a representative sample of a bearing ID shown in [figure 14](#). The top half of the image shows the smooth area that was not within the bearing/sleeve interface. The bottom of the image shows a rough, irregular and cratered appearance which was subsequently examined under higher magnification as shown in [figure 15](#). At 500x magnification, the bearing exhibited signs of extensive melting and cratering, with beads of metal adhering to the surface. Similar findings on the sleeve OD were found, as shown in [figures 16 and 17](#). EDS probes were taken in each of these key areas, with the resulting spectrograms shown in [figures 18 and 19](#). No foreign contaminants were found with these EDS probes.

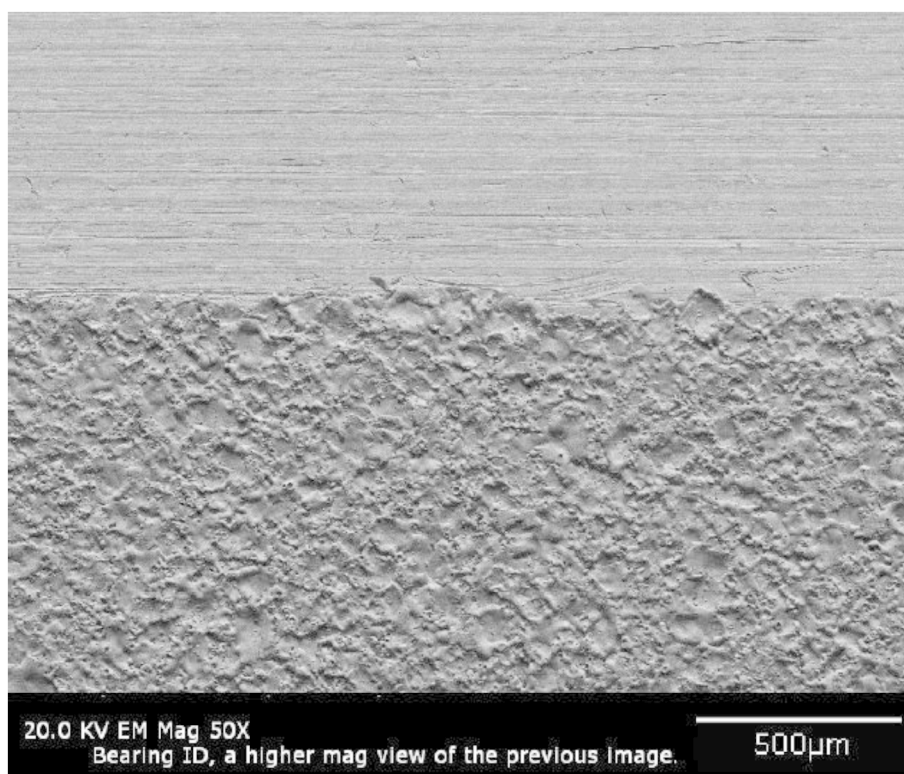


Figure 14—50x magnification of a typical bearing I.D. showing smooth and pitted areas

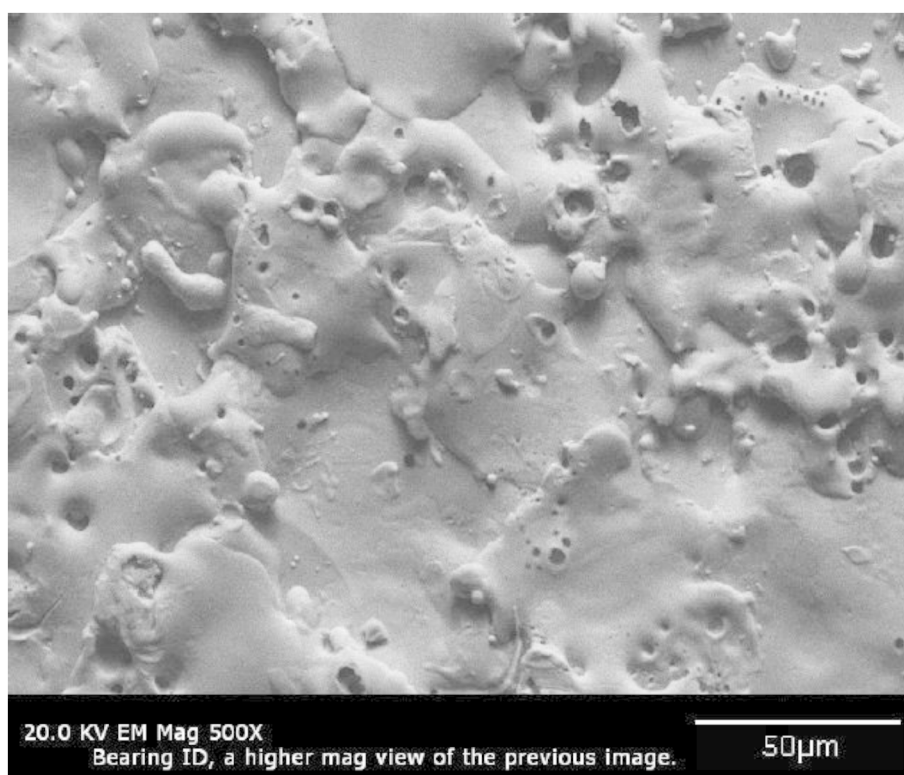


Figure 15—500x magnification of the distressed area in Figure 14

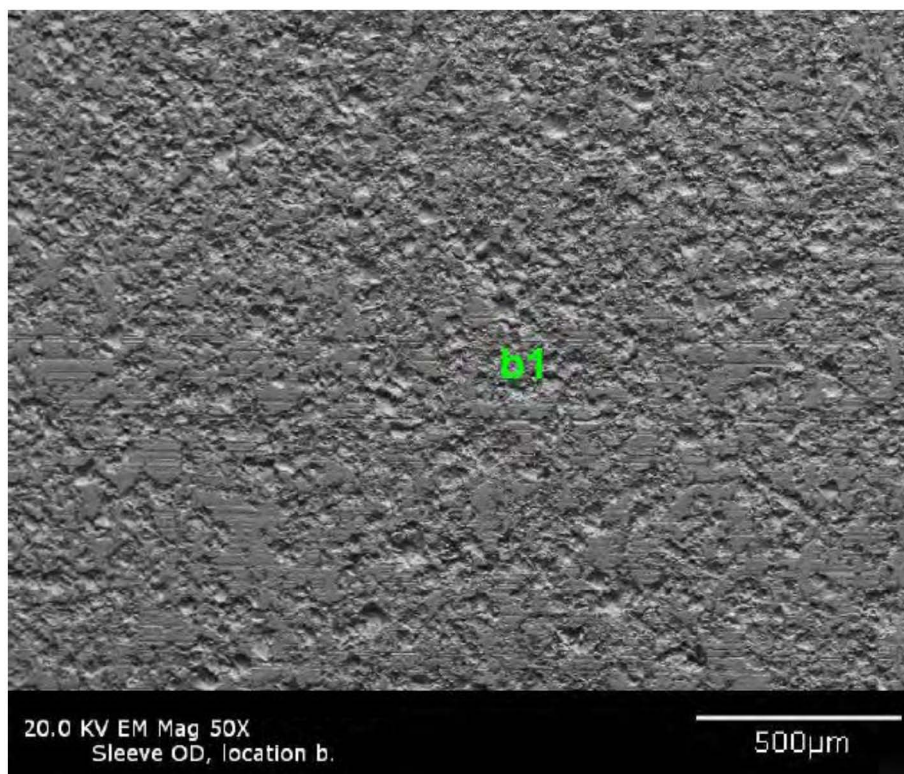


Figure 16—50x magnification of a typical bearing O.D. showing pitted areas

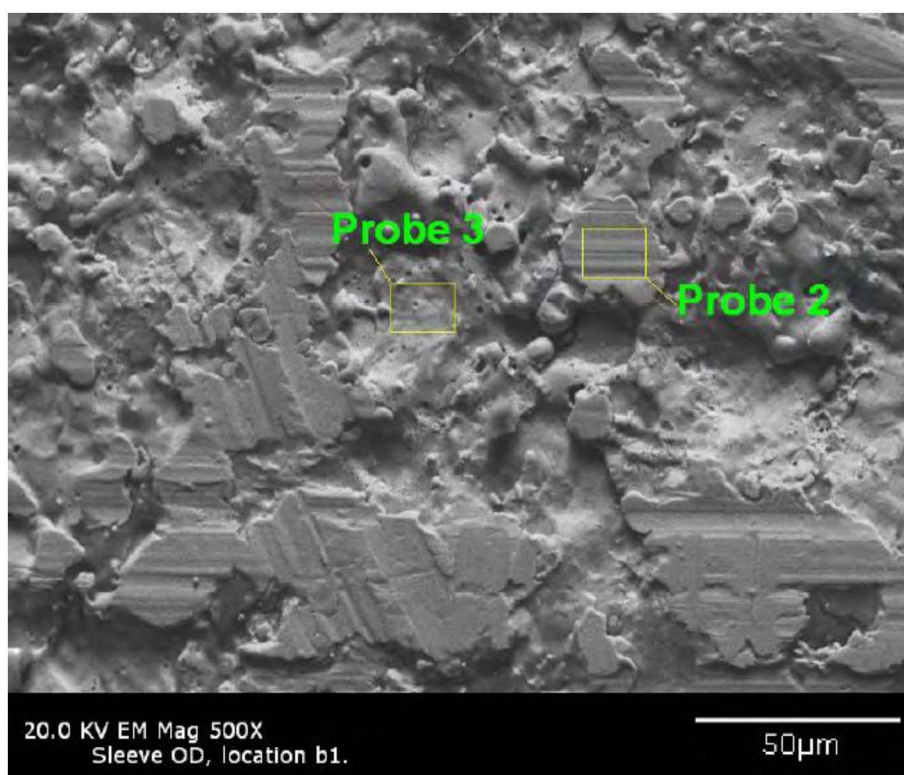


Figure 17—500x magnification of distressed area B1 in Figure 16

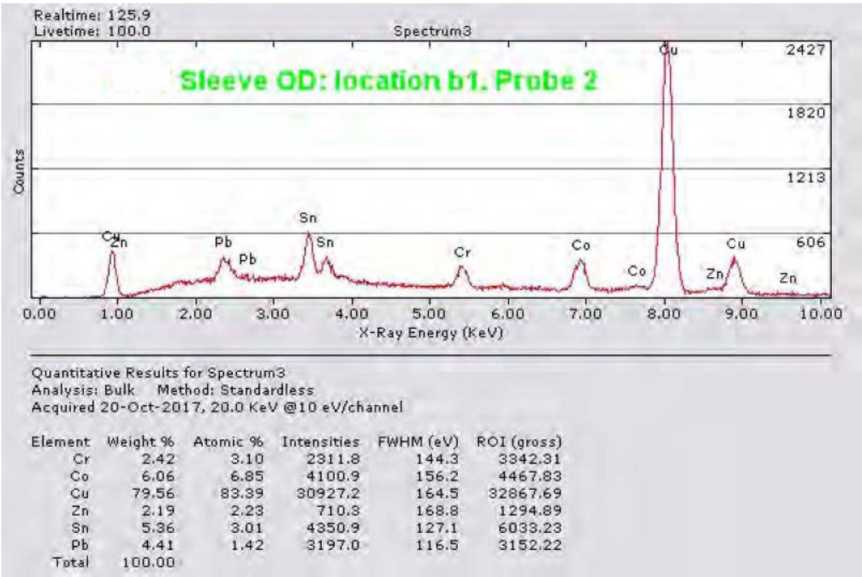


Figure 18—Spectrograph of sleeve OD: Location B1, Probe 2 as shown in Figure 18

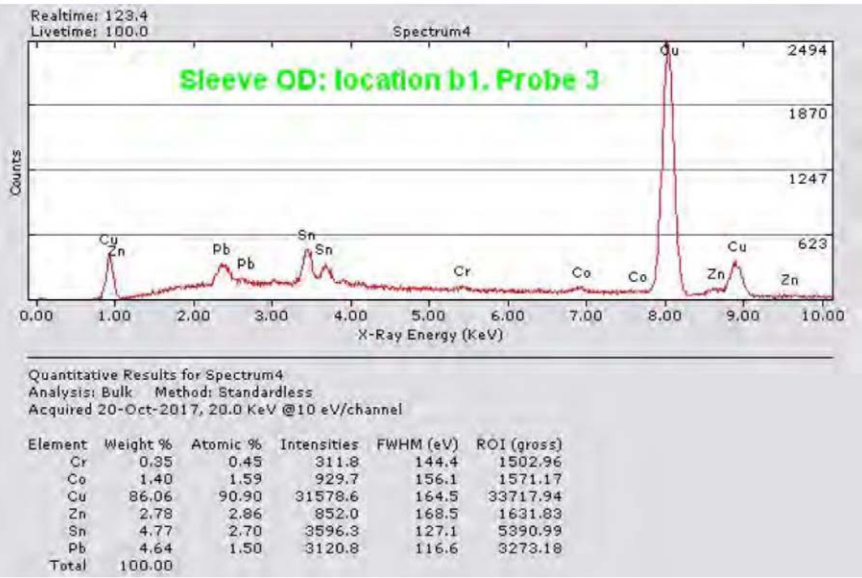


Figure 19—Spectrograph of Sleeve OD: Location B1, Probe 3 as shown in Figure 18

Only the most distressed bearings indicated fractures and cracks, however the fracture surfaces did not indicate mechanical or corrosion fatigue (progressive cracks). All fracture surfaces that were found indicated that they were of a fast overload type, with one darker, progressive crack found likely originating from an area of the bearing with minimal wall thickness.

Electrical Testing

Figure 20 shows voltage and current waveforms collected during Test 1 without any filters installed. Figures 21 and 22 show the voltage and current harmonic content measured during this test, respectively. Figure 23 shows voltage and current waveforms collected during Test 2 with only the sinewave filter installed. Figures 24 and 25 show the voltage and current harmonic content measured on the output of the sinewave filter, respectively. Finally, Figure 26 shows voltage waveforms collected during Test 3 with the sinewave and common mode filter installed. Figure 27 shows the voltage and current waveforms on the output of the common mode filter, representing the remaining high frequency content absorbed into the common mode

filter, thus prevented from flowing to ESP equipment downhole. Additionally, Figure 28 and 29 show the voltage and current harmonic content measured on the output of the common mode filter.

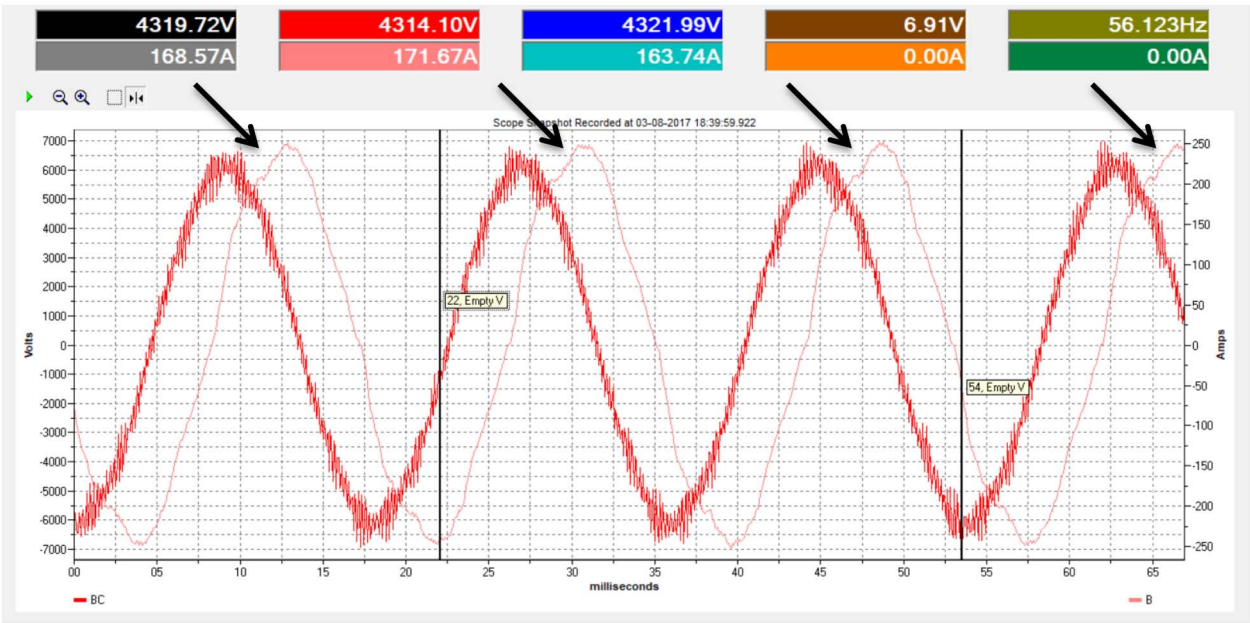


Figure 20—Test 1 (no filters) B-phase voltage and current waveforms. Note the "tilt" in the waveform as highlighted by the black arrows.

Voltage	AN	BN	CN	NG	
V RMS	2,390.8	2,405.3	2,703.6	6.9	
V PK	4,045.5	3,771.6	4,498.6	18.0	
V CF	1.7	1.6	1.7	2.6	
% THD	3.0	2.3	2.4	26.2	
Freq	56.1				
Current	A	B	C	N	G
A RMS	168.6	171.7	163.7	0.0	0.0
A PK	249.8	252.6	243.5	0.0	0.0
A CF	1.5	1.5	1.5	1.4	1.4
% THD	5.7	6.0	5.5	--	
% TDD	--	--	--	--	
K Factor	1.2	1.2	1.2		
Power	A	B	C	Total	
kW	289.193	339.606	336.818	965.617	
kVA	403.023	412.924	442.689	1,259.546	
kvar	280.705	234.890	287.275	808.727	
PF	0.718	0.822	0.761	0.767	
DPF	0.727	0.827	0.768	0.776	
	Lag	Lag	Lag		

Figure 21—Test 1 primary electrical characteristics

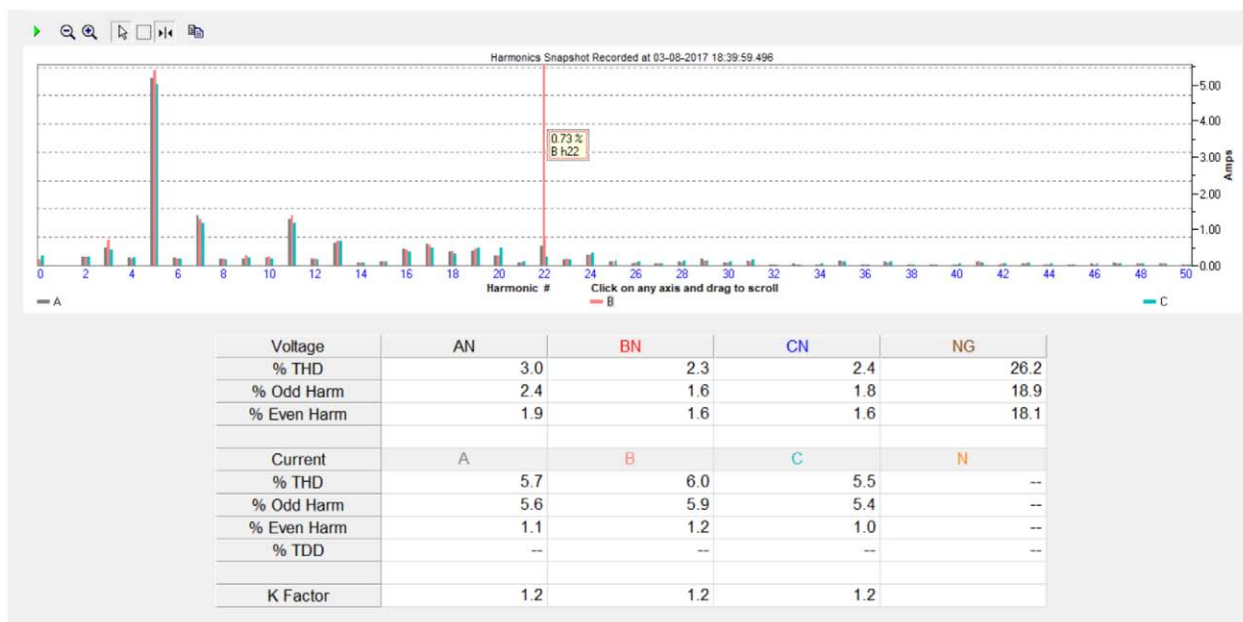


Figure 22—Test 1 spectral plot and summary of harmonic content

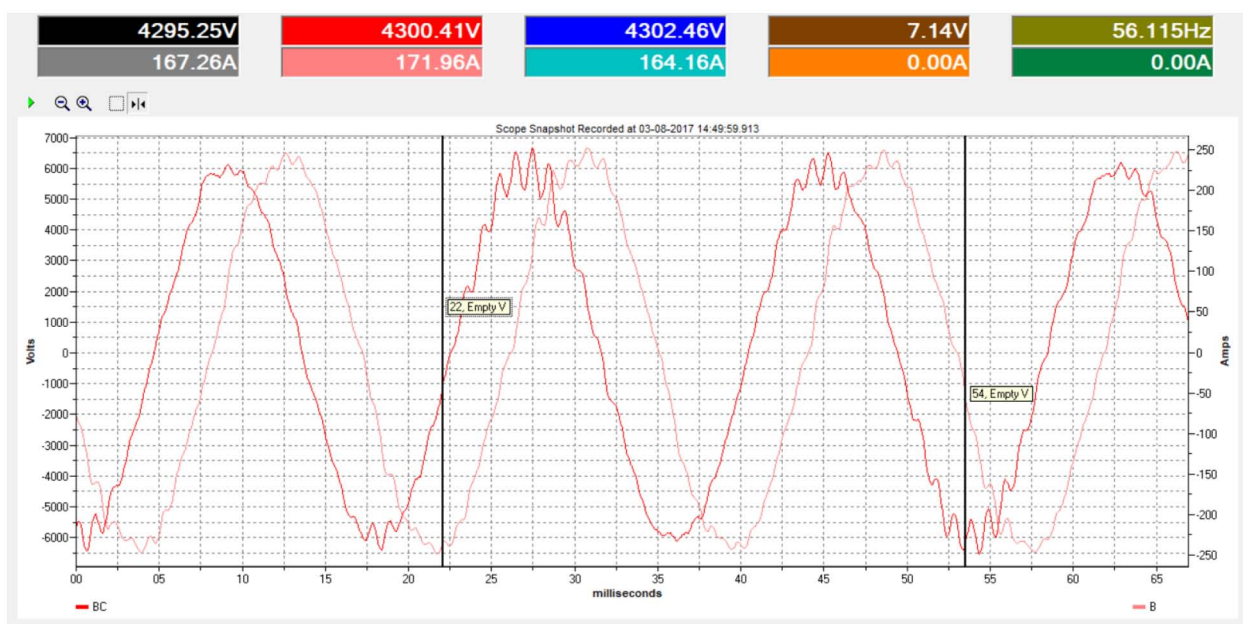


Figure 23—Test 2 (SWF only) B-phase voltage and current waveforms

Voltage	AN	BN	CN	NG	
V RMS	2,370.1	2,378.9	2,716.5	7.1	
V PK	3,970.4	3,672.2	4,357.4	14.9	
V CF	1.7	1.5	1.6	2.1	
% THD	7.0	4.2	5.8	30.1	
Freq	56.1				
Current	A	B	C	N	G
A RMS	167.3	172.0	164.2	0.0	0.0
A PK	244.3	251.5	244.6	0.0	0.0
A CF	1.5	1.5	1.5	1.2	1.3
% THD	4.8	4.7	4.7	--	
% TDD	--	--	--	--	
K Factor	1.3	1.4	1.4		
Power	A	B	C	Total	
kW	283.518	340.186	341.276	964.980	
kVA	396.431	409.065	445.928	1,252.621	
kvar	277.082	227.173	287.024	798.669	
PF	0.715	0.832	0.765	0.770	
DPF	0.718	0.833	0.767	0.775	
	Lag	Lag	Lag		

Figure 24—Test 2 (SWF only) primary electrical characteristics

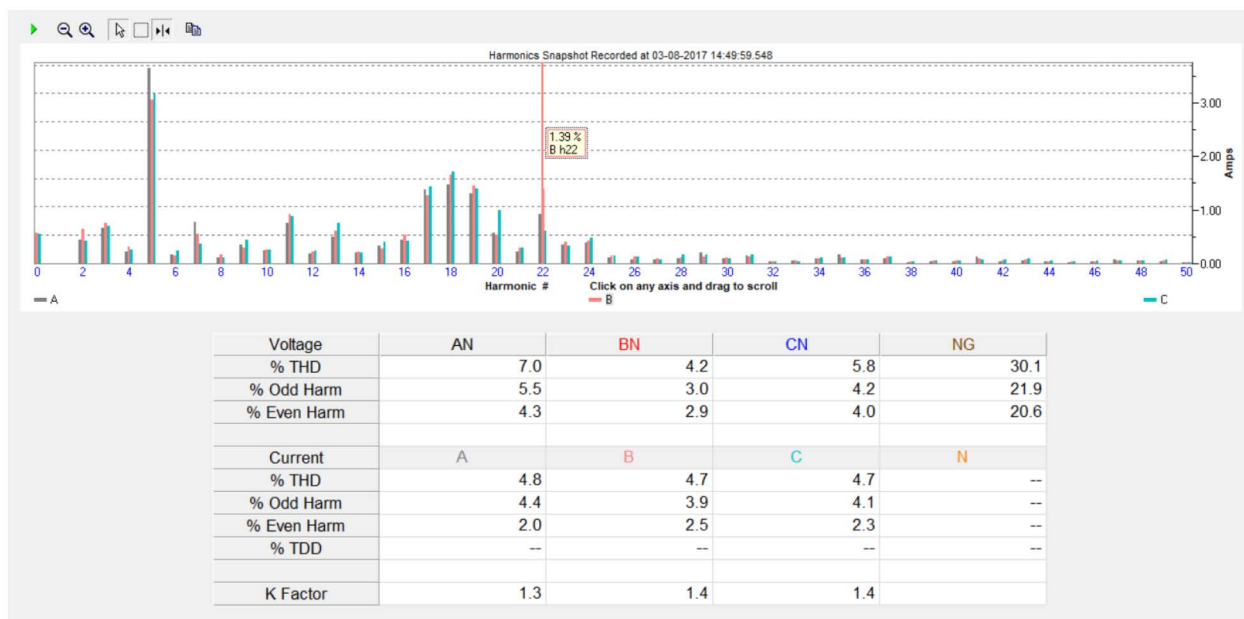


Figure 25—Test 2 (SWF only) spectral plot and summary of harmonic content

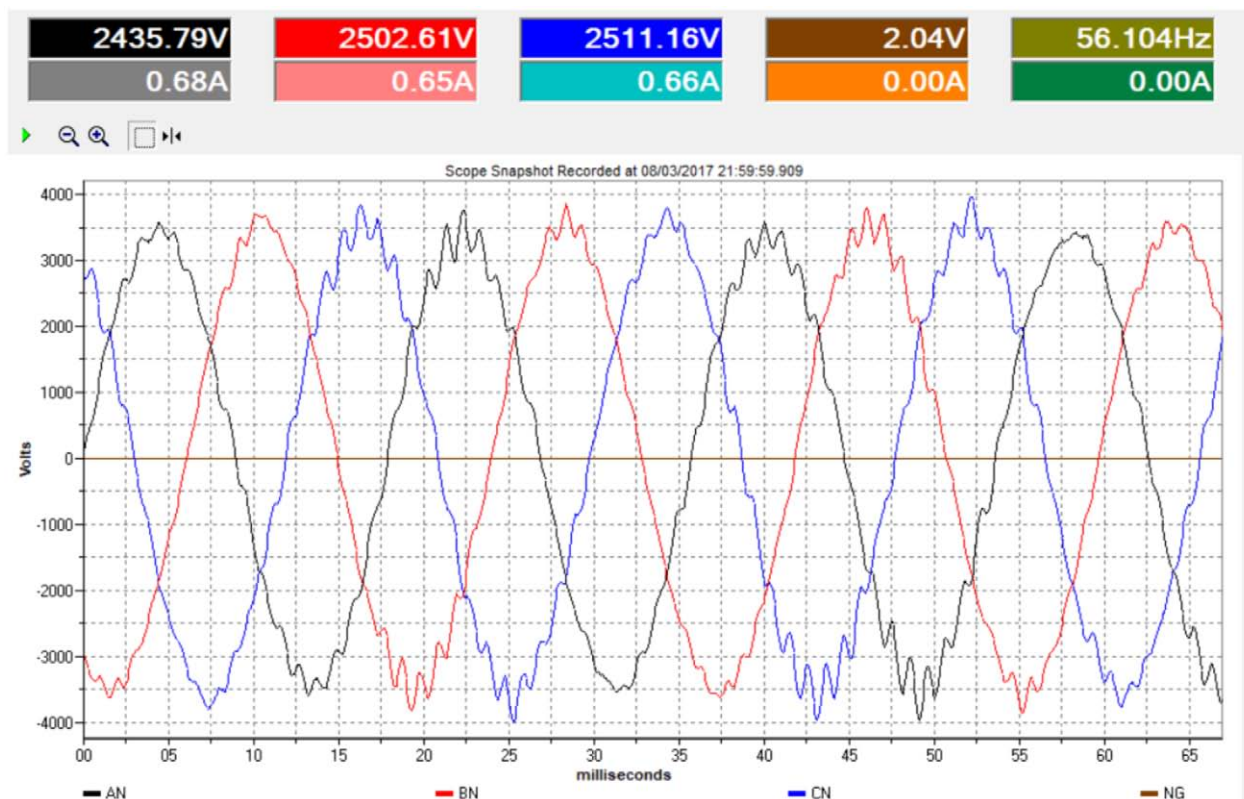


Figure 26—Test 3 (SWF & CMF filters installed) three-phase voltage waveforms

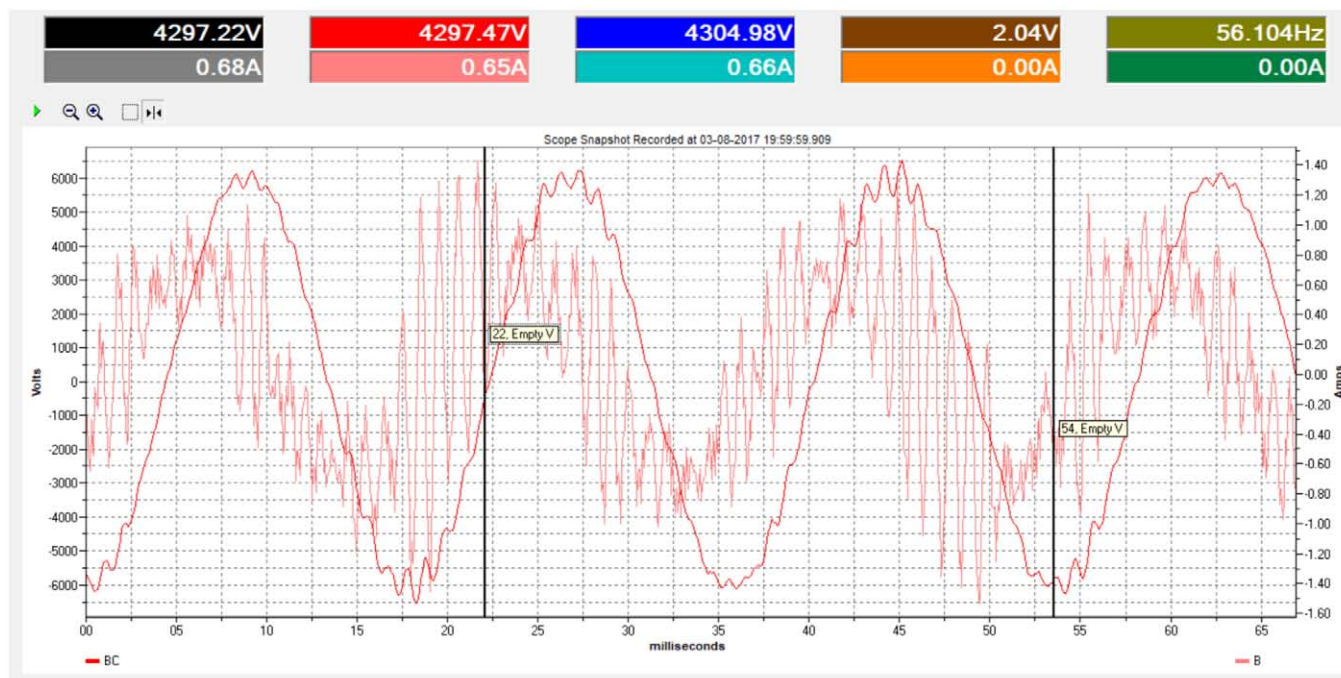


Figure 27—Test 3 (SWF & CMF filters installed) B-phase V_{out} and I_{CMF} waveforms from the Common Mode Filter, showing the high frequency content that was absorbed and shunted to ground.

Voltage	AN	BN	CN	NG	
V RMS	2,435.8	2,502.6	2,511.2	2.0	
V PK	3,836.0	3,967.6	4,053.8	4.6	
V CF	1.6	1.6	1.6	2.3	
% THD	6.2	5.1	5.9	--	
Freq	56.1				
Current	A	B	C	N	G
A RMS	0.7	0.6	0.7	0.0	0.0
A PK	1.6	1.6	1.7	0.0	0.0
A CF	2.4	2.5	2.6	1.4	1.4
% THD	99.0	84.3	88.0	--	
% TDD	--	--	--	--	
K Factor	151.3	135.6	138.7		
Power	A	B	C	Total	
kW	0.016	-0.061	-0.006	-0.051	
kVA	1.652	1.622	1.664	4.940	
kvar	1.652	1.621	1.664	4.940	
PF	0.009	0.037	0.004	0.010	
DPF	0.020	0.055	0.013	0.047	
	Lag	Lead	Lead		

Figure 28—Test 3 (SWF & CMF filters installed) V_{out} and I_{CMF} electrical characteristics from the Common Mode Filter

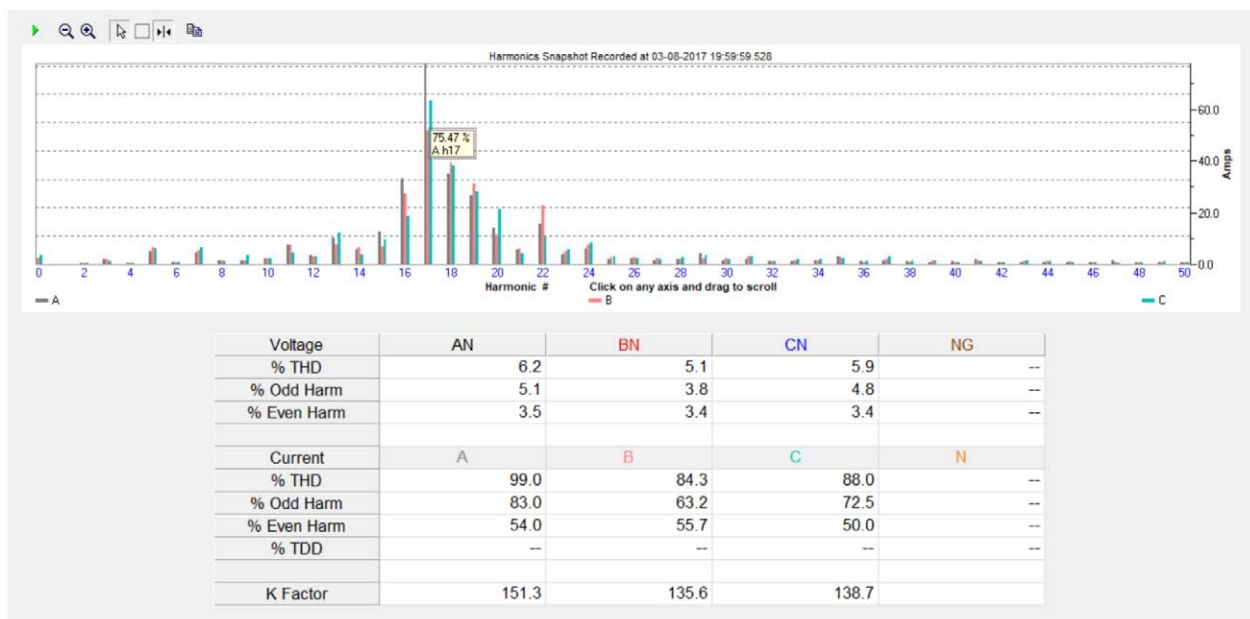


Figure 29—Test 3 (SWF & CMF filters installed) spectral plot and summary of V_{out} and I_{CMF} harmonic content from the Common Mode Filter

Analysis

Motor Teardown and Inspection

The results of the inspections indicated that the failure was most likely electrical in nature, as no mechanical process could have created the frosted appearance within the bearing interface. Furthermore, the copper plated appearance throughout the motors suggested that electrical potentials existed within the motor, creating a situation functionally electroplating components within an electrolyte. The obvious source of the copper was from the shorting bars at the end of each rotor, which are immediately adjacent to the plated bearings. Since clean ESP oil is a good dielectric but was the only fluid present, a means to conduct electricity throughout the motor needed to be found. The failure of dielectric tests proved the oil was no

longer a good dielectric however, so contamination was the only likely cause. The brass colored particles found in the oil was easily identified as being primarily sleeve material caused by the breakdown of the sleeve surface due to electrical discharge.

A serial defect of the motor design was ruled out, as the authors had significant experience running similar equipment in different wells and on different power sources (Line powered, soft starters, and other VSD's). Therefore more detailed inspections of the balance of the ESP system needed to be completed in order to fully understand the root cause of failure, including laboratory inspection of the bearings, and testing of the electrical system as well.

Bearing Lab Inspection

Since the bearing and sleeve materials were deemed appropriate and typical for a bearing application, that there were no anomalies noted in the structure of the material, there was no evidence of fatigue related mechanical failure, and that no foreign contaminants were identified, the laboratory concluded that there were no bearing manufacturing issues that could have contributed to the premature failure of the bearings.

The oil samples did not show any evidence of foreign contamination, degradation or evidence of wear. Therefore a failure in the lubricants chosen for these applications was not determined to be a causal factor of failure by the laboratory. The carboxylic ester found on the bearings was likely a lubricant applied to the bearings, and was deemed nonconsequential as well.

In all four cases, the laboratory concluded that the bearing and sleeve surfaces exhibited pitting and surface features consistent with Electrical Discharge Damage (EDD). They based their conclusion upon several findings including: the melted surface appearance of the bearing interfaces, the mixture of chrome, cobalt and sleeve elements found in the sleeve and bearings, EDS probes indicating re-melted sleeve material due to the lack of tin and zinc, and areas of the distressed surfaces showing deposition of melted materials. All of these factors are typical of an EDD condition. Furthermore, the test laboratory concluded their written report with the following findings:

"Note that "flutin" generally has a very regular machined appearance, and is often seen in association with continuous manufacturing processes that are operated at one speed for several hours at a time. "Frostin" is a special case of "flutin". The electrical bearing damage failure is relatively a new class of bearing failure and is associated with "shaft current"s which flow as a result of shaft-to-frame electrical potentials. A major source of the "shaft current" in this class of bearing failure is the potential induced between the rotating element and static element in variable speed electric motors. This potential is associated with the use of solidstate gating devices used to generate DC current for DC motors or AC variable frequency (VFD) power for AC motors". and "During virtually every VFD cycle, induced shaft voltage discharges from the motor shaft to the frame via the bearings, leaving a small fusion crater in the bearing race. These discharges are so frequent that before long the entire bearing race becomes marked with countless pits known as frosting. As damage continues, the frosting increases, eventually leading to noisy bearings and bearing failure". [Confidential company report]

Electrical Test 1

As shown in representative figures 20 through 22, this system without any filters shows a very typical PWM voltage and current waveform of a multilevel medium voltage VSD. Three phase voltage differentials are often in excess of 700V as represented by the spiked "hedgehog" like appearance. These spikes ride upon the fundamental frequency quite closely, and are representative of the high frequency content discussed at length throughout this paper. The current waveform also shows an interesting characteristic of a "notch" or "tilt" in the waveform, as shown by the arrow in figure 20. This has been attributed to the design of the motor, as tests of similar sized ESP's from other manufacturers on the same equipment does not show this attribute. This notch is a 5th harmonic, which does not contribute to any of the high frequency phenomenon discussed.

The voltage and current harmonic profiles collected are of key interest, as it allows for the quantification of values at each harmonic number. As shown in the figures, all harmonics are represented at relatively low percentages of the fundamental. This alone could seem negligible, however the resultant sum of these higher order harmonics can be quite damaging. This reality is made quite apparent in these large ESP systems, as the operating voltage and current are quite high for ESP's; 4300V L-L at 170A. Thus, a harmonic content of .5% of the fundamental would result in a potential of 21.5V at .85A. Utilizing the Yokogawa scope, the resultant amperage and percentage of the fundamental was displayed, making this relationship clear. For example at the 35th harmonic, a .23A current was measured, which was .148% of the fundamental operating current. Similarly high currents were found beyond the 20th harmonic, as represented by an unfiltered current total harmonic distortion of 5.690%.

Electrical Test 2

The addition of the sinewave filter has, through simple inspection of figures 23–25, shown that the majority of the high frequency content has been eliminated. There are ripples in the voltage waveform, especially evident at the peaks, which are caused by the sinewave filter. Sinewave filters, like all filters, tend to amplify certain harmonics which are represented by the ripples shown here. The current waveforms also show improvement, with less high frequency content on the waveform and the 5th order notch largely removed. The harmonic analysis verifies what the inspection of the waveform has indicated. We see an increase of the 17th harmonic relative to Test 1. The 17th harmonic measured here is not high enough in magnitude to cause significant capacitive coupling, and thus it is not able to contribute to the high frequency failure modes described previously. Of particular note however, is the significant reduction of harmonic content beyond the 20th harmonic, in most cases with the content approaching 0% of the fundamental. Thus we can see that the addition of a sinewave filter has reduced the electrical content that is an antecedent to high frequency electrical failure modes. The current total harmonic distortion has been reduced in this case to 4.466%, or 1.224% lower than Test 1.

Electrical Test 3

Again, through simple inspection of figures 26–29, it becomes clear that the inclusion of the common mode filter on this system results in essentially all of the high frequency content being removed from the voltage and current waveforms. While it's rather difficult to see directly from the power cable waveforms, it becomes quite clear when inspecting the output current into the common mode filter. Here we see the common mode filter is removing nearly .7A of high frequency content. Previous research has shown that high frequency content of this magnitude can cause bearing fluting in similar systems, thus it is reasonable to assume it can do so here as well [5]. In any case, the elimination of high frequency electrical content should at very least reduce motor and cable heating, thus improving the life expectancy of the system. Of final note, inspection of the voltage and current waveforms shows that they closely approximate a pure sinewave (as shown in Figure 2), therefore the supply power is more similar to grid power than found in Tests 1 & 2.

Correlation of Airborne Ultrasound to Electrical Testing

During VSD/ESP system testing in August 2017, ultrasonic condition monitoring equipment was utilized at the wellhead to check for any electrical or safety issues. The device was set to 40 kilohertz (kHz) which is appropriate for detecting airborne ultrasonic emissions. The wellhead junction box and surface cabling were then inspected by the authors. The equipment was scanned three times; once without any filters installed, again with only a sine wave filter installed, and finally with the sine wave and common mode filters installed. During these inspections, no standard electrical failure modes were detected (arcing, tracking, corona, or mechanical looseness). It is important to note that when performing ultrasonic inspection of electrical equipment, the decibel reading is considered irrelevant in and of itself, but the relative sound can be used to gauge magnitudes and severity of electrical phenomena heard (including the aforementioned arcing,

tracking, corona and mechanical looseness). Therefore, the simple presence of an unusual sound is enough to indicate a problem. In the first test case with only a sine wave filter, there was what is best described as "inverter noise" emanating from the armored cable between the wellhead junction box and the wellhead penetrator. When the same scan without a sine wave filter installed was performed, the amount of noise was noticeably more pronounced. Finally, with the sinewave filter and common mode filter installed, there was no detectable sound anywhere on the surface cabling. These tests were run during the same afternoon, and no significant weather changes occurred which could have compromised these findings.

This series of tests served to indicate that in certain circumstances, airborne ultrasound could be used to detect the high frequency component of the electrical waveform on a VSD/ESP system. Further, it indicated that the high frequency electrical component of the waveform was reduced by the sine wave filter, and was practically eliminated when using both the sine wave filter and common mode filter. Since these measurements were taken near the wellhead penetrator, the best conclusion is that the motor is no longer subject to the high frequencies induced by the inverter when properly sized sine wave and common mode filters are installed.

The authors work to date suggests that the ultrasonic probe is detecting the ionization of gas molecules due to the high frequency electrical leakage through the cable insulation to ground. The ionization results in ultrasonic turbulence which is transmitted through the air, and is detected by the sensor on the probe. The creation of ultrasound via ionizing gases is the same principle used to detect arcing, tracking and corona, and is a well-documented and accepted means to detect electrical faults [1]. This phenomenon has been tested by the authors at numerous wells throughout the country, both in Geothermal and Oil applications, with similar results. Therefore, it is believed that ultrasound, used correctly, can at very least qualitatively identify high risk assets, such that appropriate electrical filters can be proactively installed. Finally, it appears likely this technique can be used as a simple means to verify the installed filters are functioning correctly while performing standard electrical ultrasonic inspection rounds on the ESP system. This area has proven to be an area of research interest, as the collection and analysis of ultrasonic data is relatively easy. Characterization of waveforms may also provide insights to other aspects of the ESP, all without shutdowns or use of more specialized tools.

Discussion and Conclusion

Since all of the motors from the two separate fields exhibited very consistent physical evidence during teardowns, it became very evident system commonalities needed to be more closely investigated. Since the authors had significant experience working with both the ESP and VSD equipment manufacturers, it was deemed unlikely that there was an inherent design flaw in any particular component. Both manufacturers were reputable and known for providing good quality equipment. This was proven through the laboratory inspection of the bearings, indicating they were of expected quality and design. Furthermore, since laboratory inspection of the bearings eliminated all nonelectrical failure modes from the discussion, only the electrical components of the ESP system needed to be understood to determine the root cause. Even in the case of the cable failure, it was determined that skin effect heating and heat generated by reflections of the high frequency content were the most likely cause of failure. This excessive skin effect heat caused by high frequency electrical content in an already hot environment resulted in breakdown and failure of the splice, and caused the cabled insulation to lose adhesion to the conductor.

Subsequent electrical tests of these systems further indicated that significant high frequency harmonics existed due to system design. Further it was proven that without filters, the original design did not mitigate harmful high frequency harmonics prior to the motor. The addition of a well-designed sinewave filter did in all test cases significantly reduce the high frequency content measured at the wellhead, but not enough to fully mitigate the possible failure modes. This was reinforced during Test 3, where .66 amps of high

frequency harmonics were shunted to ground through the common mode filter; a current magnitude that has been demonstrated to cause similar failures on the similar equipment in offshore ESP systems.

After inspecting over 20 motors all exhibiting the same key evidence, the following sequence of events was deemed the most probable failure progression:

1. Due to the tight clearances within a bearing, shaft currents were circulating through the bearings which were caused by the high frequency switching of the VSD.
2. The electric discharge across the bearings resulted in surface imperfections, increasing bearing heat and releasing metallic debris into the oil.
3. This metallic debris reduced the dielectric strength of the oil, allowing the electrical discharge within the bearings to occur more frequently.
 - a. Copper electroplating represents a parallel path through the oil for the high frequency currents, thus the copper plated the lower electrical potential bearing materials.
4. As the oil became increasingly more contaminated, the lubricity of the oil was reduced, further increasing the heat generated within the bearing.
5. Materials in the oil begin scoring the bearing surfaces, further increasing heat and causing physical damage to the sleeves (the softest material in the bearing).
6. Due to excessive heat and mechanical damage, a bearing fails, scattering significant debris throughout the oil. Higher weight particles travel down towards the bottom of the motor due to gravity, however the motors continue to run in this condition due to the high horsepower.
7. Subsequent bearing failures occur due to the increased contamination, further reducing the lubricity of the oil until the rotor is locked to the stator or the stator is physically damaged, both resulting in an instantaneous overcurrent trip of the VSD.
8. In some cases, electrical cable failures will occur prior to motor failures. This is most often caused by skin effect heating and ring wave reflections which result in cable insulation and splice failures.

Since the root cause of the failures in these scenarios appears to have been poor power quality, the addition of filters should improve mean times between failures (MTBF) of the equipment, and eliminate the failure modes most often associated with high frequency electrical signals. Upon installation of sinewave and common mode filters, the MTBF for these ESP systems has extended from as little as 8 months, to the same wells now approaching two years of continuous operation. Considering the equipment is otherwise the same as during the rash of failures during 2015, the addition of filters has demonstrably improved equipment run life in a significant way.

Recommendations and Lessons Learned

Since ESP motors and electrical cable are not, and cannot easily be optimized for VSD operation, the only solution for ESP operators and system designers is to correct the power quality issues before systems fail. Therefore in all VSD powered ESP applications, it is recommended that a cost/benefit analysis be performed, where the expected lifecycle cost of a system with sinewave and common mode filters is weighed against the expected lifecycle cost of the system without filters. If filters are deemed economically beneficial, they need to be appropriately designed by an electrical engineer with demonstrated technical knowledge of the entire ESP system, along with deep understanding of high frequency electrical harmonics, their effects, and means of mitigation.

In unconventional wells utilizing VSD/ESP systems, the selection of sinewave filters becomes particularly critical. The wide operating ranges demanded by production engineers results in situations where sinewave filters will be ineffective at best, and harmful at worst. By the nature of their design constraints, sine wave filters are effective for only specific frequency ranges. When an ESP system

is operated well outside those design constraints (a function of the system capacitance and inductance primarily), the sine wave filter ceases to function appropriately. In extreme cases, the system could enter into electrical resonance resulting in system failure.

While this paper is technical in nature, it's important to note that the marginal cost of adding filters to ESP systems is insignificant when considering the benefits gained from additional system up time and reduced replacement costs. It is strongly suggested that ESP operators carefully consider the "hidden costs" as well, such as, decreased safety performance, procurement costs, collateral damage, costs for motor tear downs and inspections, opportunity costs, and the cost multipliers associated with emergency and reactive work. When analyzed closely, it's reasonably shown that the total cost to a business for a single ESP failure can be substantially higher than the simple cost of equipment replacement and lost production. Analyzed in this way, the inclusion of sinewave and common mode filters becomes negligible when amortized over the entire life cycle cost of the well. In short; improving ESP reliability now will cost less than numerous unnecessary failures that will most likely result from poor power quality later.

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